Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 01

Subject: BSC102/AP **Date: 03/09/2026**

CO1: To provide students with a basic understanding of laser operation

Q. No.	Questions	Marks
1	In a Ruby laser, chromium ions (Cr^{3+}) undergo radiative decay between an excited metastable state E_2 and the ground state E_1. The energy difference between these two laser levels is $\Delta E = 1.79$ eV. (a) Calculate the frequency (ν) of the emitted radiation. (b) Determine the wavelength (λ) of the emitted laser beam in nanometers (nm) and state which region of the electromagnetic spectrum it occupies.	**2**
2	A Helium-Neon (He-Ne) gas laser produces a highly coherent beam through stimulated emission at a characteristic wavelength of $\lambda = 632.8$ nm. 1. Find the transition frequency (ν) of the emitted photons. 2. Calculate the energy difference ($E_2 - E_1$) between the upper and lower lasing levels in both Joules (J) and electron-volts (eV).	**2**
3	Consider a two-level atomic system with an energy separation corresponding to a transition wavelength $\lambda = 694.3$ nm (Ruby laser). 1. Calculate the ratio of the population of the upper level (N_2) to the lower level (N_1) at thermal equilibrium at room temperature ($T = 300$ K). 2. At what hypothetical temperature T would the thermal population of the upper level reach 10% of the ground state population ($N_2/N_1 = 0.1$)?	**4**
4	In a gas laser medium at thermodynamic equilibrium at a temperature of $T = 500$ K, the lower energy level is chosen as the reference state ($E_1 = 0$ eV) with population N_1. The population of an excited level E_2 is measured to be $N_2 = 1.25 \times 10^{-4}\, N_1$. 1. Using the Boltzmann distribution $N(E) = N_0 \exp(-E/k_B T)$, calculate the energy of the upper level E_2 in electron-volts (eV).	**2**

NB: Experiment is the only means of knowledge at our disposal. Everything else is poetry, imagination. *– Max Planck*